DEEPAK KUMAR NITHIYANANDAN

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WORK EXPERIENCE

Data Scientist Intern – QuerKey Inc., California — ***June 2025 – March 2026***

- Implemented and processed 100+ hours Python-based video analysis pipeline using OpenCV and Pandas to process large volumes of hospital CCTV footage. Extracted frame-level data, generated structured JSON outputs, and embedded precise timestamps to track ambulance arrival and departure events.
- Authored and optimized SQL queries for RAG, designing efficient indexing strategies to speed up query execution. Validated retrieval accuracy against hospital records to ensure LLMs received the correct context for downstream question-answering tasks.
- Automated the end-to-end Superset chart creation process through REST APIs and Microsoft power automate by dynamically configuring chart payloads, metrics, and filters.
- Built and deployed a Streamlit hospital analytics dashboard with global filters that synchronize across all charts. Integrated an AI-driven chart recommendation system where LLMs suggested the best visualization type, significantly reducing user effort and decision support for hospital administrators.

Graduate Research Assistant - New Jersey Institute of Technology — ***May 2025 – August 2025***

- Collaborated with Dr. Roni Barak Ventura on analyzing 20+ years of firearm legislation impacts using causal inference and long-term time series datasets covering background checks and firearm related homicides.
- Applied information-theoretic methods such as entropy and mutual information and delayed mutual information to identify causal links between background checks and firearm homicides on 10K+ records.
- Developed a moving window framework in Python to capture evolving causal trends across two decades of firearm data, allowing detection of periods where laws strengthened or weakened the observed relationships.
- Managed data preprocessing and visualization of firearm-related background checks, homicide fractions, legislative timelines and improved analysis accuracy.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, C, Spark, JavaScript, HTML, CSS, Bash, Solidity, MATLAB.
- **Databases:** MySQL, PostgreSQL, MongoDB, Apache HBase, NoSQL, LanceDB.
- **Data Analytics:** Looker, Sigma, Statistical modelling, A/B Testing, Data Mining, KPI Reporting, peoplesoft.
- **Data Visualization:** Matplotlib, Seaborn, Plotly, Superset (REST APIs), Streamlit, Power BI, Tableau.
- **Cloud & Libraries:** Pandas, NumPy, SciPy, Scikit-learn, PyTorch, OpenCV, Mediapipe, Hadoop, PySpark, MapReduce, Oozie, Git, Excel, SHAP, Databricks, Snowflake, GCS.

ACADEMIC PROJECT

Flight Data Analysis — ***NJIT***

- Engineered a data processing pipeline to analyze 22 years of U.S flight records stored in HDFS.
- Developed Java Based Hadoop MapReduce Jobs to aggregate flight delays, cancellations, airline performances.
- Designed a Mapper, Reducer and data cleaning logic to process large scale structured data across multiple nodes.
- Automated and coordinated MapReduce jobs using Apache Oozie workflows, reducing manual execution.
- Validated job outputs and generated results for downstream reporting and visualization.

  Tech Stack: *Java, Hadoop, MapReduce, HDFS, Apache Oozie.*

EDUCATION

New Jersey Institute of Technology — ***Newark, NJ***

MS in Data Science (Computational Track) — *Graduation Date: May 2026*

- *Activities/Awards: Entrepreneurial Experience Badge, Research Assistant*

PES University — ***Bangalore, India***

B. Tech in Computer Science and Engineering — *Graduation Date: May 2024*

- *Activities/Awards: DAC scholarship recipient, Best Capstone Project*